

MovableBag: Exploring Asymmetric Interaction for Multi-user Exergame in Extended Reality

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ABSTRACT

Previous research shows that virtual reality (VR) exergames motivate users' physical activity with immersion and enjoyment. However, it limits the movement for players to perform intense exercises in a small space with a complex tracking setup. In addition, asymmetric interaction of multi-user exergame in the same physical space is also lacking in the discussion in extended reality (XR). So, we present MovableBag, a substitutional robot that can be adapted to multi-user exergames with prop. It also serves as an agent to allow users from the physical world to interact with the VR user to become an XR application. Additionally, a preliminary study has been conducted, and a multi-user boxing exergame is designed to investigate the usage of the work. It also informs future design for more use cases in XR for asymmetric interaction between VR and non-VR users.

CCS CONCEPTS

• Human-centered computing → Haptic devices; • Hardware → Emerging interfaces.

KEYWORDS

Multi-user Exergame, Asymmetric Interaction, Substitutional Robot, Encountered-type Haptic, Extended Reality

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1 INTRODUCTION

In the previous generations, playing video games was considered a sedentary activity. With technological advancement, game consoles with tracking ability, such as the Nintendo Wii, enable interaction by body movements, followed by more consoles such as PlayStation Move, Microsoft Kinect, and Switch. These give rise to exergames

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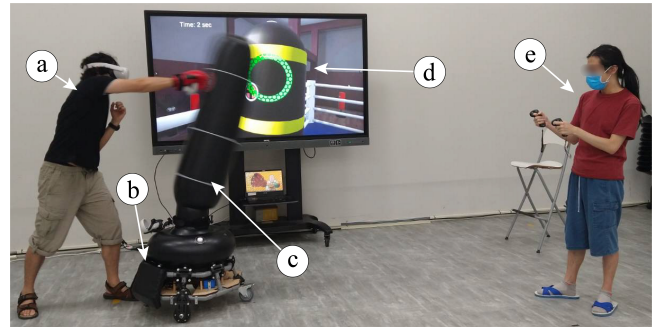


Figure 1: Multi-user boxing exergame with MovableBag: (a)Player(boxer), (b)Basement, (c)Punching airbag(Target props), (d)Player's view, (e)coach(Coach)

to motivate players to exercise while playing by engaging multiple muscles instead of only hands compared to traditional games. Meanwhile, some research shows that adaptation of VR technology to exergame can have a significant impact in terms of motivation, performance, presence, embodiment, and participation in the player's experience and thus beneficial for exercise and rehabilitation. [1] However, there are some limitations to the VR experience for player movement in the physical world, including (i) potential dangers of movement while wearing the VR headset, (ii) the cable for some HMD and trackers that restrict players' movement, (iii) the timely detection of boundaries in the virtual environment (VE) that affect player experience, (iv) the interaction with a physical prop in the physical world that might pose a risk to players during movement, and (v) the movement simulation in which the opponent or target is not shown in the physical world.

Apart from the above issues, there have been some researches investigating the multi-user interaction in exergame with different approaches, but mostly about player-to-player interaction. Furthermore, the asymmetric interaction between the player and the coach is less explored. Therefore, we present MovableBag, a substitutional robot for multi-user exergame, which serves as an interaction medium for target tracking and force absorption from the player's punch, and also an agent for the coach to interact with the player in the extended reality(XR) environment by including both player and coach in same physical space. In this paper, we develop a multi-user boxing exergame for the preliminary study to investigate the usage of MovableBag on immersion and experience via asymmetric interaction in XR and how it impacts the experience of the exergame.

2 RELATED WORK

This work aims to improve the exergame experience by providing a wider variety of haptics and exertion to increase immersion in the virtual world, and to allow multi-user interaction of the exergame in the XR environment.

2.1 Exergame in Virtual Reality

There have been some investigations showing the benefit of VR exergames. In "Race Yourself" [5], the players rode a static bicycle while wearing a VR HMD. Their performance data, such as exertion, were recorded. This information was then used to create virtual avatars that represented the previous performance of the players. The novelty of this approach allowed the researchers to identify that the sense of competition can greatly influence the intrinsic motivation of players to win a race, and performance improvement was identified. But have not yet discussed the competition among players, or the coach involvement.

The haptic devices have also been used in the VR exergame to increase immersion by stimulating different senses or force for the interaction with the users' weight and movement. Pull-Ups [7] implemented a haptic pull-up system that provided kinesthetic force feedback to players using a couple of specially designed straps attached to the ceiling. This haptic device created forces that pulled the player, depending on his weight and resistance strength. With this suspension system, users could enjoy different activities that targeted multiple body parts, such as arms and shoulders, in interaction with their weight. The result shows that the system improves realism and enjoyment, but the system is limited to small-area movement and upper-body exercise.

The use of physical props for VR experiences and games has also been investigated. These props, equipped with trackers, can represent different objects in the virtual world, thus increasing the realism of VR games. Props can serve as an environmental aid in VE as visual cues and adapt to user actions. The interaction with props can be classified into either passive haptics, in which the player applies force over the props, or active haptics, in which the prop exerts force towards the player with technology. Additionally, some research has explored the possibilities of prop mobility in the physical world, which can be tracked in VE. RoomShift [6] adopted a swarm robot with an optical tracking system to create encountered-typed haptic feedback in VE corresponding to the movement of the props in the physical world. However, the proposed system requires a complex setup and can only handle limited weights; for example, a user cannot sit on the device or carry heavy objects to different locations. These limit the fierce interaction in the VR exergame.

2.2 Asymmetric Interaction in Extended Reality

Asymmetric Interaction in XR is another critical issue for multiple users with varying roles, especially when the VR user is wearing an HMD. It is challenging to be aware of the physical environment and communicate with external users. There are discussions about adopting different user interfaces for asymmetric interaction between users in the shared environment. [8] design a climbing system that involves bystanders in interaction with VR users using a VR controller. ShareVR[4] uses position tracking and floor

projection technology to include spectators as part of the VR experience for VR users. While these approaches offer more possibilities for enabling multiple users using different interfaces to interact with each other in the same shared space, there is still room for exploration in terms of haptic feedback and mobility of the props incurred in the VR experience, which may further extend the application of asymmetric interaction for exergames requiring more fierce movement and tracking need.

3 DESIGN CONSIDERATIONS

To enhance player experiences in the VR exergame, we aggregated the consideration of the interaction between the player and the coach in the same shared space before developing the system. From the player side, the VR experience should be realistic, adapt to the physical environment, such as the size of the room or the furniture, and consistent with the characteristics of the sports equipment [3]. Therefore, it was important to define the basic movement of the exercise that a player would perform in a limited space safely and efficiently with the aid of sports equipment. As mentioned in [2], people trigger multiple muscle groups that increase the effectiveness and impact of exercise and health in general when interacting with the yoga ball (balance ball). Therefore, we are inspired to design a device capable of carrying different sports props. We generate the following design considerations that should be taken into account.

- (1) Ability to support strong forces and weight, including force from the user and heavy props.
- (2) Ability to move around to allow the player to have a wider range of movement to enhance the workout experience.
- (3) The prop on the device can provide encountered-typed haptic feedback for user interactions, such as touching or punching.

From the coach side, his role is to facilitate player's training in the physical environment by controlling the device movement and to instruct player's movement during the training session.

4 DESIGN AND IMPLEMENTATION

MovableBag consists of the base structure with the steering system, integrated with a software solution that includes the tracking system to develop a multi-user boxing experience for user study.

4.1 Base Structure and Steering System

As shown in Figure 2a, the base structure is designed in an octagonal shape of two tiers for stability and is made of plastic tubes. The connector parts were designed and created by a 3D printer. The hollow in the center of the device is designed to hold large exercise props, such as a punching bag or a dummy target. The bumper cushions were designed to fit the structure for safety reasons. The steering system consists of caster wheels and motor wheels, enabling the device's mobility. The two motor wheels are separately controlled by the hub motor driver and powered by Li-ion battery. (Figure 2b) The initial design used six caster wheels and two motor wheels for greater stability. However, it was found that the system would move erratically with six caster wheels due to reorientation of the wheels after tests. Therefore, it is modified to a set of only two caster wheels and two motor wheels for smoother movement

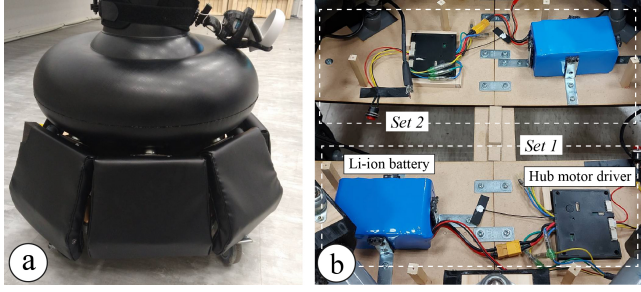


Figure 2: Structure of the Base:(a) Overview, (b) steering system

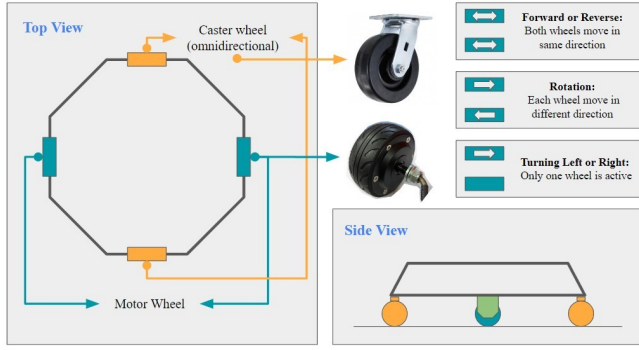


Figure 3: MovableBag's steering system model

and is also capable of carrying the punching bag. (Figure 3) According to the specification, the motor wheels offer a top speed of 18km/h, with a power adjustment function that includes 25%, 50%, 75%, and 100%, and a remote controller controls each wheel. (Figure 2b) The straight-line speed test is conducted indoors with a 7.9 m flat floor with different levels of power adjustment as a pilot study to evaluate device mobility. The results of the average speed tested are 1.06m/s, 2.01m/s, 3.61m/s, 3.72m/s, respectively, without loading. This research used the minimum power (25%) due to safety concerns and the weight-bearing requirement.

4.2 Tracking System

For the tracking system, we adapt the wireless headset Oculus Quest 2, which allows players to move around with the ease of setup. Our system uses the VR controller for tracking our haptic device by a 3D printed controller holder, which could detect the acceleration, velocity, and location when the user's force is exerted onto the punching bag. Another VR controller is attached to the boxer glove to interact with the bag via collision detection.

4.3 Multi-user Boxing Exergame

We have developed an XR Boxing Exergame with Unity to demonstrate the functionality of our device and conduct a preliminary user study. We placed a 6kg punching bag with a VR controller on MovableBag to integrate it into VE as a virtual target. The exergame is designed for two users; the player(boxer) and the coach(coach).



Figure 4: Design for Boxing Exergame in VR: (a) Prop tracker and Hand controller, (b) Virtual target with indicator

The player wears the boxing glove with a VR controller, then the virtual glove is visualized in VE simultaneously. Each time the player punches the bag, the collision is detected by the other VR controller attached to the base of the bag. (Figure 4a) The physical impact of the virtual target in VE will be calculated based on the acceleration and velocity data from both controllers' movements. In case the player punches the bag by hand without a controller, the physical impact of the virtual target can still be simulated, though with less accuracy as there are only oscillation data from the VR controller in the basement. The coach needs to facilitate the player to play the exergame by controlling the MovableBag to move around the punching bag. The coach can decide the strategies to interact with the player; aggressively approaching the player, dodging the player's punch, etc. It is designed without in-game audio to allow room for verbal communication between the player and the coach. To maintain smooth gameplay, there is an indicator in the virtual bag (Figure 4b) to remind the player to stay close enough to the device, otherwise the headset may lose tracking of the VR controller. From our own technical test, Oculus Quest 2 HMD may lose track of trackers if they are apart beyond around 2 meters, and the indicator will turn red to remind the player to get closer to the device. Another purpose is to indicate the "target face" of the virtual target that the player should keep facing for punching.

5 USER STUDY

To validate the usability of our device, we conducted a user study with ten pairs of 9 men and 11 women (average age = 25, SD = 1.82) to experience VR boxing Exergame as "boxer" (punching the bag) and "coach" (controlling MovableBag + and guiding the boxer). The study consists of two sessions with a two-minute exergame experience in each session. The group of two participants takes turns being a boxer or a coach in each session. After two sessions, they fill out an iGroup Presence Questionnaire(IPQ), and then a short interview is conducted.

The IPQ result of boxer experience (as shown in Figure 5a) revealed that, generally, the participants as boxer felt present and immersed in the virtual world. However, the average score is lower in terms of spatial presence, involvement, and experienced realism. This might be explained by the lack of a controller for the other hand to visualize both hands, making a boxer feel less immersive. Boxer feel less confident when punching with another hand without

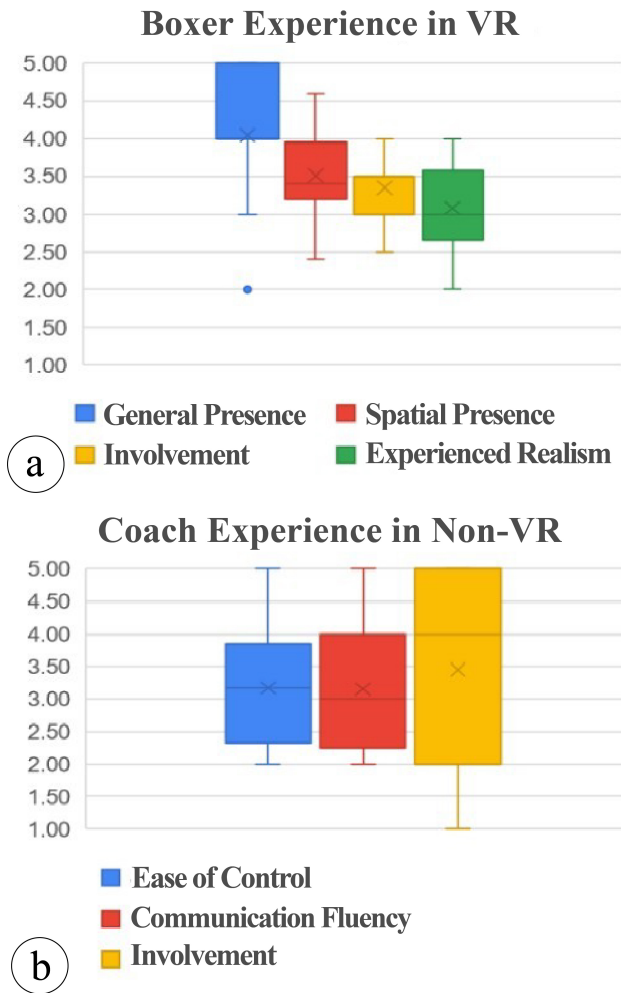


Figure 5: Questionnaire Result:(a)IPQ result of Boxer Experience in VR, (b)General feedback of Coach Experience in Non-VR

visualizing the second glove, which may negatively impact spatial presence. In addition to the IPQ, there is some qualitative feedback from the VR Boxing Exergame of participants related to tracking issues. The Base or the punching bag might physically hinder the boxer, as sometimes the tracking is not accurate enough to alert the boxer to evade. Another issue is that the VR controller in the base might be blocked by the punching bag and lose track when the boxer cannot follow the rotation of the device controlled by the coach. Although we had developed a calibration function to update the location of the virtual target when tracking is lost, the location is updated abruptly once the tracker is detected successfully. This might hurt realism, which can be revised by gradually translating to the correct position. Regarding the enjoyment of the exergame, most of the participants show that it is pleasant to be both a boxer and a coach. While the result from coach experience (as shown in Figure 5b) shows that some participants as coach found that it was not easy to control the device during the session, which might

lead to less enjoyment than being the boxer. Some participants as boxer mentioned the exciting experience with a larger available movement area in the VR Boxing Exergame.

6 DISCUSSION

From the above result, MovableBag shows some positive impact on the enjoyment of VR exergame. The "degree of movement" in our VR exergame is highly valued by the participants, which might be one key factor of consideration for designing an exergame. It also reveals the issue of safety when adopting robots in VR exergames, especially exercises that require intense body movement, such as boxing. The player might be knocked back by the punching bag and hurt if the physical and virtual targets are not highly synchronized, as the player cannot be aware of the physical target since wearing a VR headset. Another potential issue for coach is that they might mistakenly control the device, which can hurt the player or himself. Additionally, the coach controlling the device, who needs to pay close attention to the player during gameplay, might be less motivated in the game. There might be some motivation system designed in the VR exergame for the coach to have more engagement and interaction with the player in VE, such as creating shared goal tasks or competing with users.

7 CONCLUSION AND FUTURE WORK

The contribution of this work is implementing a haptic device that could explore the possibility of multi-user VR exergame with asymmetric interaction between the player and coach in the same physical space. While the player is training in the VR environment with HMD, the coach in the physical environment can control the MovableBag to interact with the player. It also sustains forces and can transit between fixed placement or moving around, opening this way to a broader range of options when designing exergame experiences. For instance, agility and strength training with the use of a punching bag is demonstrated in the user study from a collaborative perspective. Another example could be from a competitive perspective of an exergame like eSport to allow players to challenge each other. Lastly, the exergame with more players and teams can be further explored with more devices in the same physical space.

In the future, the structure of MovableBag can be improved to support heavier props by using more solid materials, such as iron or aluminum. The indoor microwave sensor would be adopted for the motion detection of MovableBag in a larger area and to release the VR controller back to the boxer's hand for dual-hand detection. In addition, more comprehensive applications other than boxing exergames with MovableBag can also be explored and discussed, for example, other combat sports or ball sports, or even extending it as an agent for collaboration between users in the VE and users in the physical world for applications other than exercise.

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REFERENCES

- [1] Felix Born, Sophie Abramowski, and Maic Masuch. 2019. Exergaming in VR: The Impact of Immersive Embodiment on Motivation, Performance, and Perceived Exertion. *2019 11th International Conference on Virtual Worlds and Games for Serious Applications (VS-Games)* (2019), 1–8.
- [2] Kathryn M Clark, Laurence E Holt, and Joy Sinyard. 2003. Electromyographic comparison of the upper and lower rectus abdominis during abdominal exercises. *Journal of strength and conditioning research* 17, 3 (August 2003), 475–483. [https://doi.org/10.1519/1533-4287\(2003\)017<0475:ecotua>2.0.co;2](https://doi.org/10.1519/1533-4287(2003)017<0475:ecotua>2.0.co;2)
- [3] Nuša Farič, Henry Potts, Adrian Hon, Lee Smith, Katie Newby, Andrew Steptoe, and Abigail Fisher. 2019. What Players of Virtual Reality Exercise Games Want: Thematic Analysis of Web-Based Reviews. *Journal of Medical Internet Research* 21 (09 2019).
- [4] Jan Gugenheimer, Evgeny Stemasov, Julian Frommel, and Enrico Rukzio. 2017. ShareVR: Enabling Co-Located Experiences for Virtual Reality between HMD and Non-HMD Users. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (*CHI '17*). Association for Computing Machinery, New York, NY, USA, 4021–4033. <https://doi.org/10.1145/3025453.3025683>
- [5] Alexander Michael and Christof Lutteroth. 2020. Race Yourselves: A Longitudinal Exploration of Self-Competition Between Past, Present, and Future Performances in a VR Exergame. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI Conference on Human Factors and Computing Systems, Vol. 2020-April)*. Association for Computing Machinery, USA United States, 1–17. <https://doi.org/10.1145/3313831.3376256>
- [6] Ryo Suzuki, Hooman Hedayati, Clement Zheng, James L. Bohn, Daniel Szafir, Ellen Yi-Luen Do, Mark D. Gross, and Daniel Leithinger. 2020. *RoomShift: Room-Scale Dynamic Haptics for VR with Furniture-Moving Swarm Robots*. Association for Computing Machinery, New York, NY, USA, 1–11. <https://doi.org/10.1145/3313831.3376523>
- [7] Yuan-Syun Ye, Hsin-Yu Chen, and Liwei Chan. 2019. Pull-Ups: Enhancing Suspension Activities in Virtual Reality with Body-Scale Kinesthetic Force Feedback. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (*UIST '19*). Association for Computing Machinery, New York, NY, USA, 791–801. <https://doi.org/10.1145/3332165.3347874>
- [8] André Zenner, Felix Kosmalla, Marco Speicher, Florian Daiber, and Antonio Krüger. 2018. A projection-based interface to involve semi-immersed users in substitutional realities. In *2018 IEEE 4th Workshop on Everyday Virtual Reality (WEVR)*.